From PRs to Performance

Using metrics and AI to coach stronger engineering teams





MOMENTUM
Developer Conference
powered by The Circuit

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Metrics should start conversations, not end careers.

01

PROBLEMS WITH METRICS

Why many scoreboards kill morale.

Why most engineering metrics fail in practice

Metrics Ignored

Dashboards exist, but leadership still decides on vibes.

Individual Scoreboards

Overcorrecting into per-engineer rankings that breed fear.

Velocity Without Context

A number you argue about in retro, not one that drives decisions.

Optimize for the Metric

Engineers game the measure instead of pursuing the outcome.

"What gets measured gets gamed." — Goodhart's Law

Where our approach fits in the landscape

We didn't invent this. We borrowed the best ideas and tuned them for coaching.

BENCHMARK

DORA

Google research · 4 metrics

— Deployment frequency
— Lead time for changes
— Change failure rate
— Time to restore

Is the delivery system healthy?

FRAMEWORK

SPACE

Microsoft research · 5 dimensions

— Satisfaction
— Performance
— Activity
— Communication
— Efficiency

How to measure without weaponizing.

THIS TALK

Our take

3 tiers · 1:1-ready

— Outcomes
— System Signals
— Behaviors

What do I say Monday morning?

A simplification of SPACE with DORA-style thinking underneath; tuned for coaching conversations.

THE FRAMEWORK

Outcomes vs. System Signals vs. Behaviors

LAGGING

Outcomes

Did we ship? Did customers adopt it? Did revenue move?

The rearview mirror.

LEADING

System Signals

Velocity trends, hours-per-point, time-to-merge, review rounds.

The dashboard gauges.

QUALITATIVE

Behaviors

Code review depth, testing habits, collaboration patterns.

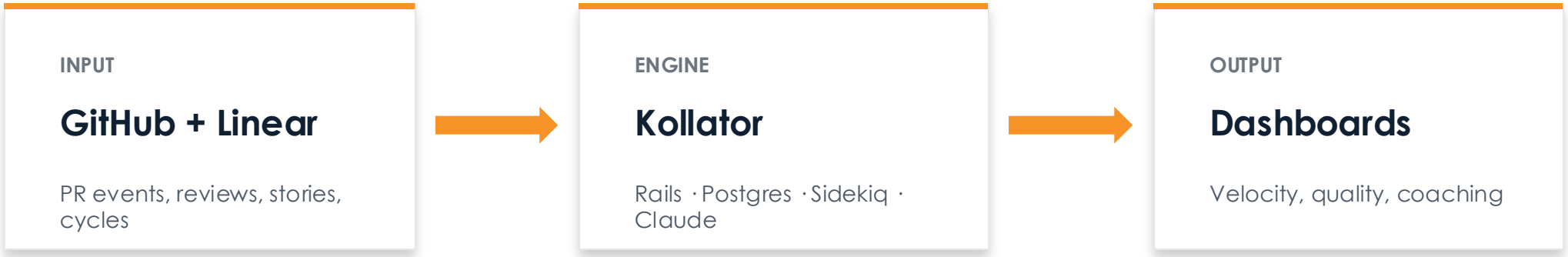
The engine light.

Most orgs only measure outcomes, then wonder why they can't course-correct.

THE DASHBOARD BLUEPRINT

What we built

Web Application, DB, Background Jobs, Charting, AI

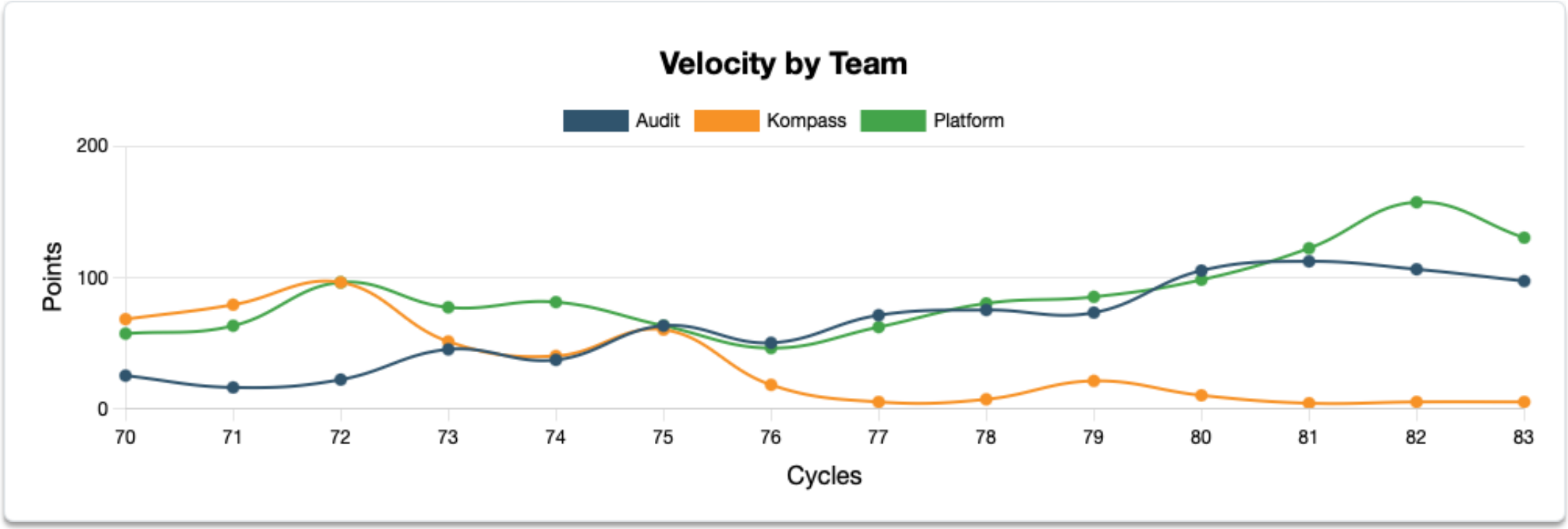


- AI layer sits ON TOP of existing data
- Weekly pipeline: classify, score, coach, analyze
- Monday morning; fresh data/analysis

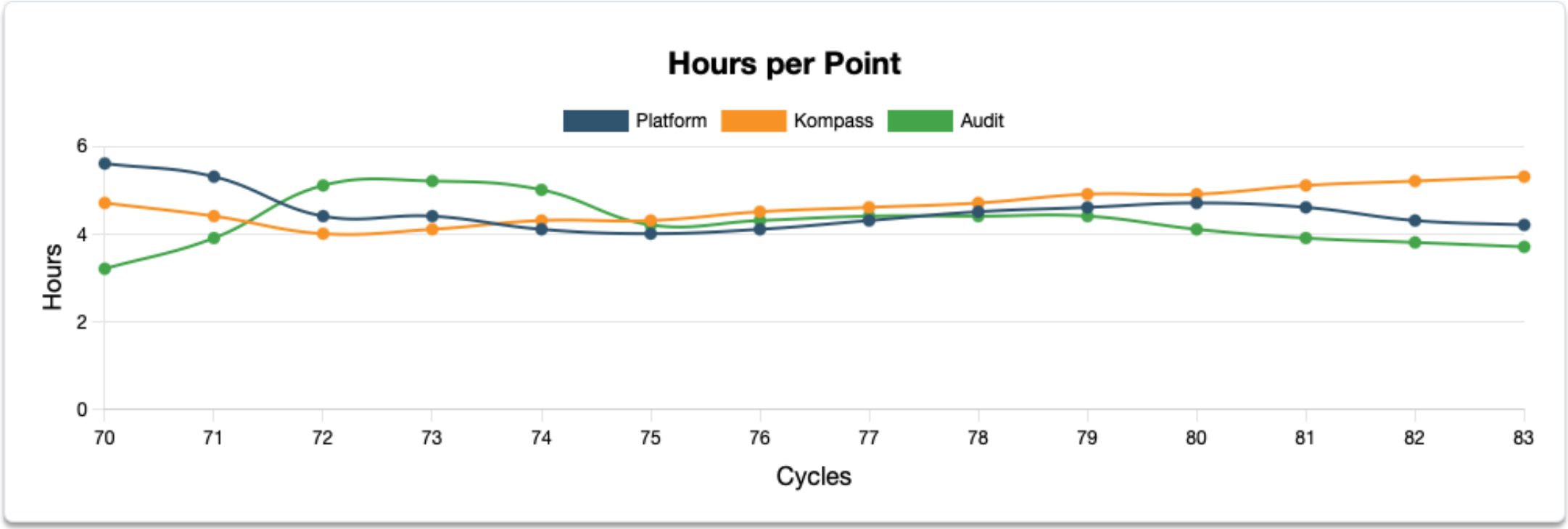
02

VELOCITY, ESTIMATION, AND FORECASTING

Team Velocity Dashboard



Hours per point ·the predictor

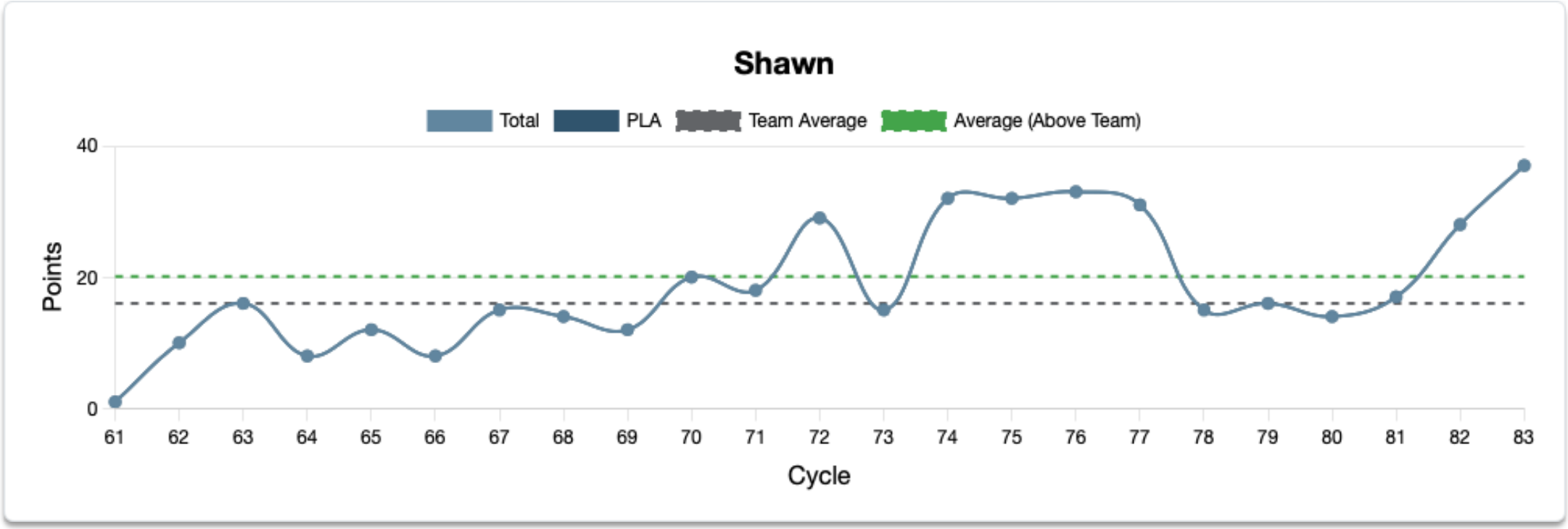


— SAME EFFORT. DIFFERENT CALIBRATION.

Raw velocity hides this. Hours per point exposes it and stops the "that team ships more" argument dead.

Trending up? Estimation is drifting. Trending down? Getting efficient or cutting corners. Check quality.

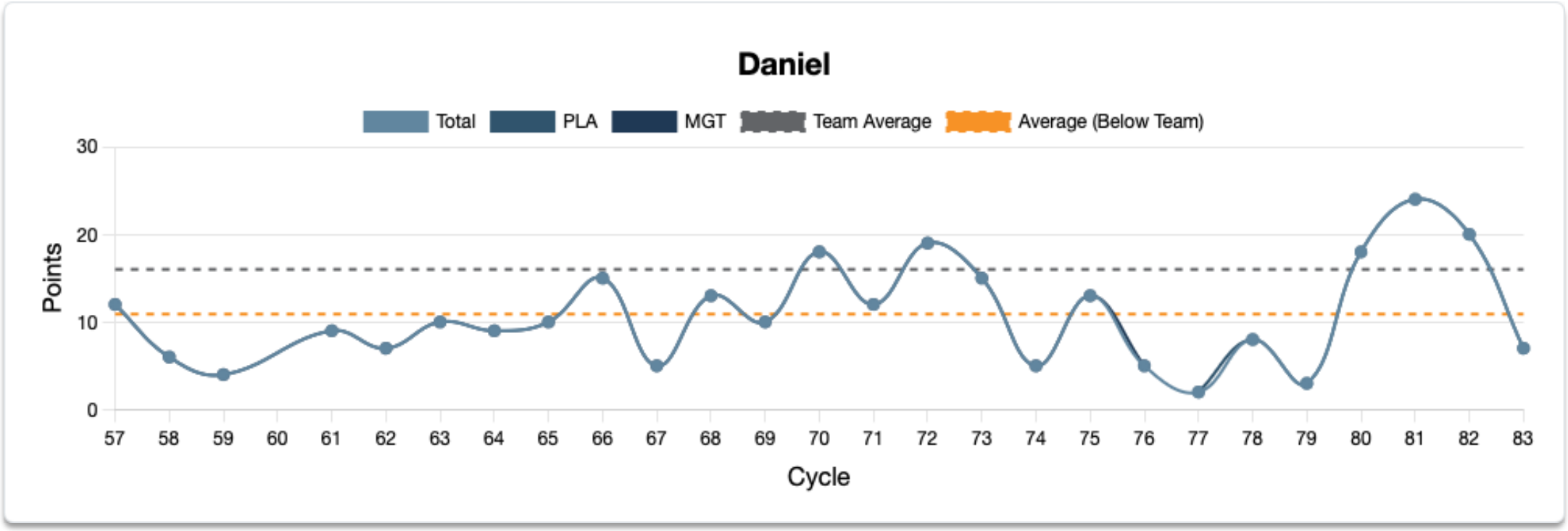
Individual Velocity · No Leaderboards



The question isn't "who's fastest?" It's "is this pattern changing, and should I ask about it?"

Team average line for context · no ranking · a conversation starter, not a verdict.

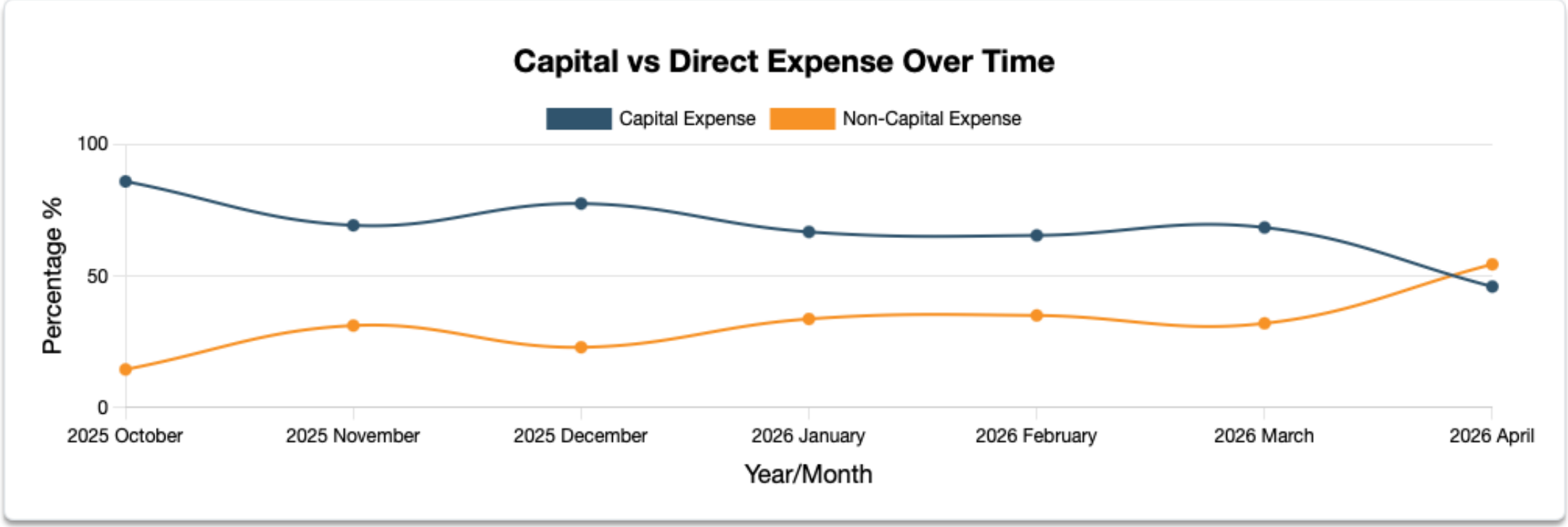
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Capital vs Expense

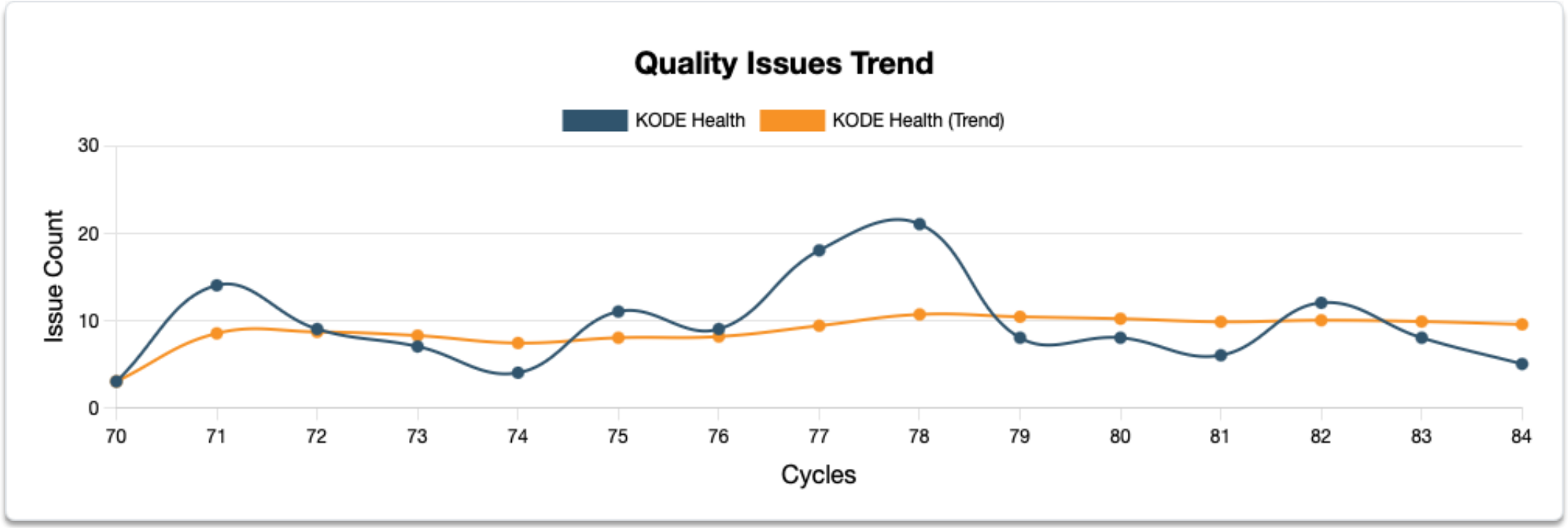


03

SIGNALS THAT REVEAL QUALITY AND RISK

Not what shipped · how it got there

Quality Trends



Quality AI Analysis

— Mixed Generated 1 day ago

Reactive work dropped sharply after a rough Cycle 78, but a spike in Cycle 82 shows we haven't fully stabilized — the team is still getting pulled off planned work every other sprint.

Cycle 78 was an outlier at 21 quality issues, largely driven by a burst of triage across all three sub-teams during what looks like a major feature rollout (audit programs, feature flags, integration wiring). The team cut that roughly in half and held steady for three cycles (6-8 issues), which is real progress. However, Cycle 82 jumped back to 12 — mostly on the Platform (PLA) team — with a mix of data integrity bugs, UX issues, and a production email leak, suggesting new feature work is shipping without enough safeguards.

Suggested Action

Pick the two or three areas generating repeat issues — Calendly integration, feature flag gating, and LOS/chart workflows — and schedule a focused hardening block in the next cycle. Even half a sprint dedicated to adding guard rails in these specific areas would likely prevent multiple interruptions in the cycles that follow.

Quality AI Analysis

★ Bright Spots

The team drove reactive work down from 21 issues in Cycle 78 to a low of 6 in Cycle 81 — that's a meaningful improvement and shows the team can stabilize when they have space to do it.

AUD and MGT sub-teams have nearly disappeared from the quality issue list since Cycle 78, suggesting those areas have been cleaned up effectively.

The ratio of triage-to-bug shifted over time — later cycles show more items correctly labeled as bugs, which suggests the team is getting better at classifying and understanding what's coming in rather than just firefighting.

⚠ Watch Items

Platform (PLA) team is absorbing nearly all reactive work

When one team handles 80% of interruptions, their planned feature delivery becomes unpredictable. This makes it hard to give Product reliable commitments on roadmap items.

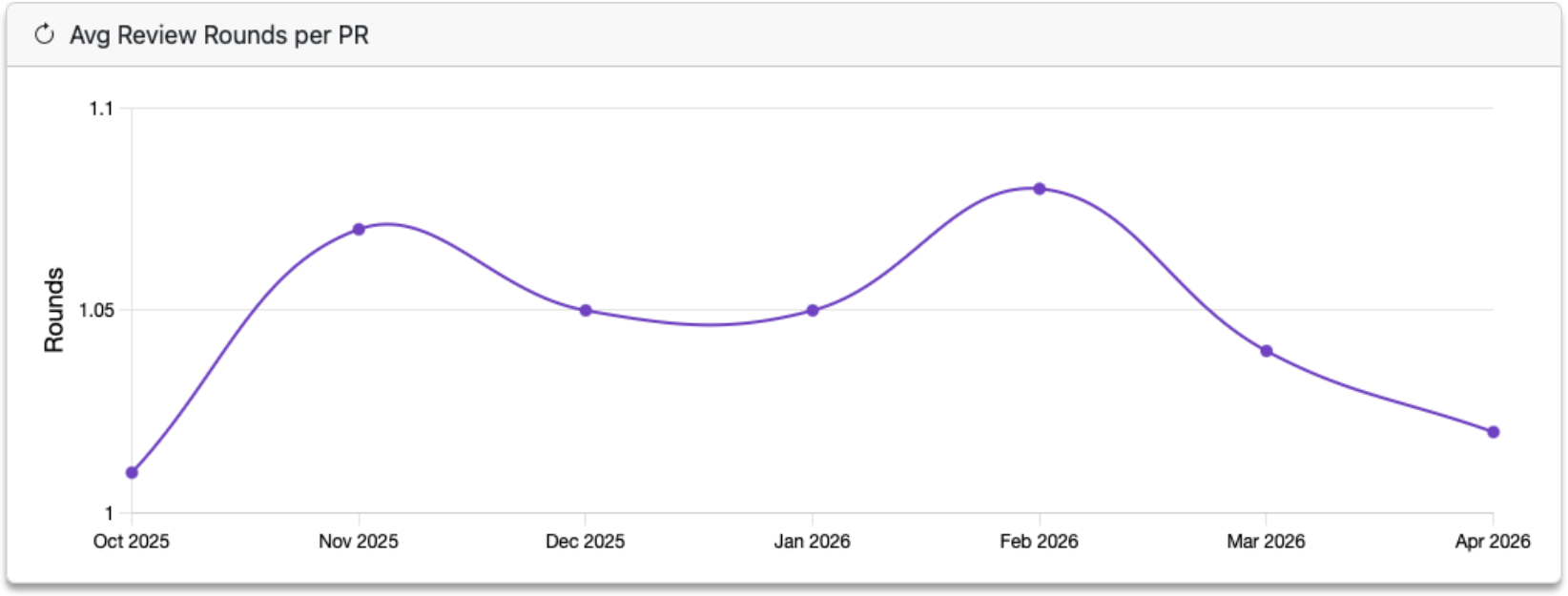
Data integrity and environment isolation issues surfacing in production

Cycle 82 included staging emails reaching business users and FK violations on real workflows. These erode trust with internal users and clients — and they're the kind of issues that escalate quickly.

Feature flag discipline — features leaking into production before they're ready

Cycle 83's 'Training Overhaul visible in prod' and earlier gating issues suggest flags aren't being consistently managed. Unfinished features showing up in production creates confusion for users and support load for the team.

Review & rework patterns



— AVG REVIEW ROUNDS

Back-and-forth cycles per PR.

— TIME TO MERGE

PR open → merged.

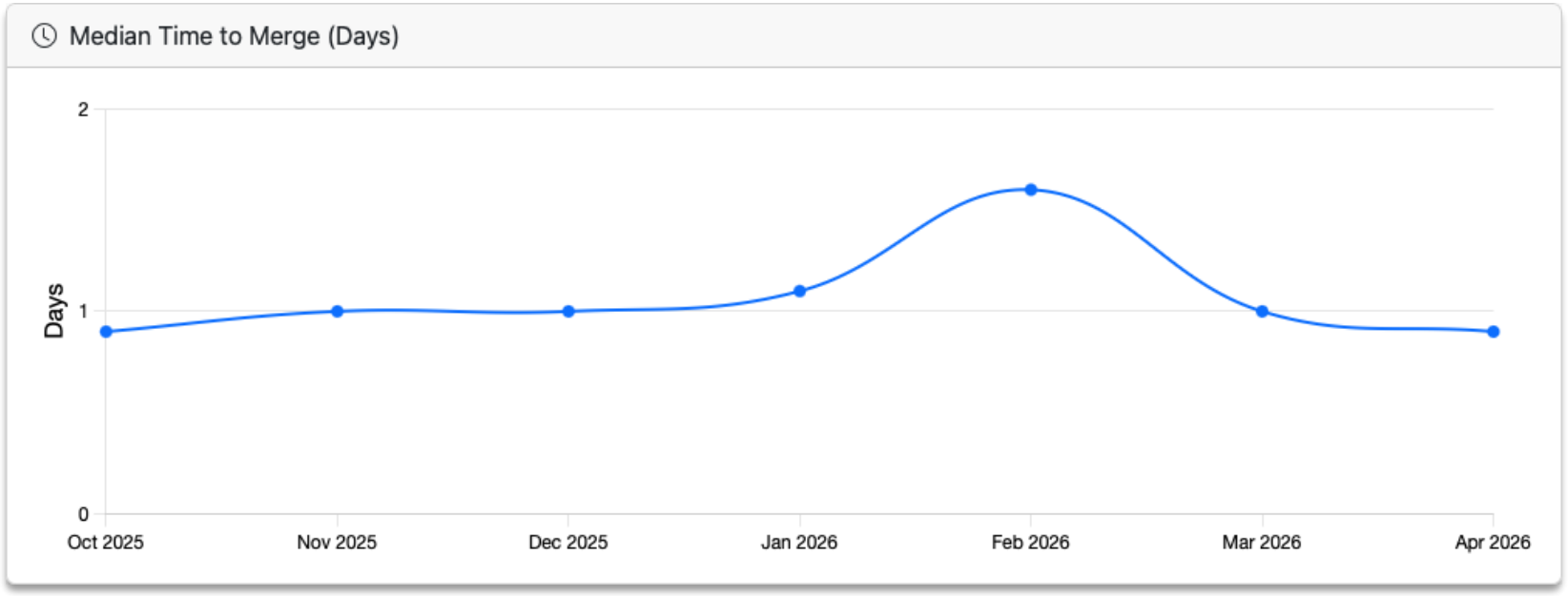
— CHANGES REQUESTED

% of PRs pushed back.

— TIME TO FIRST REVIEW

Collaboration health.

Review & rework patterns



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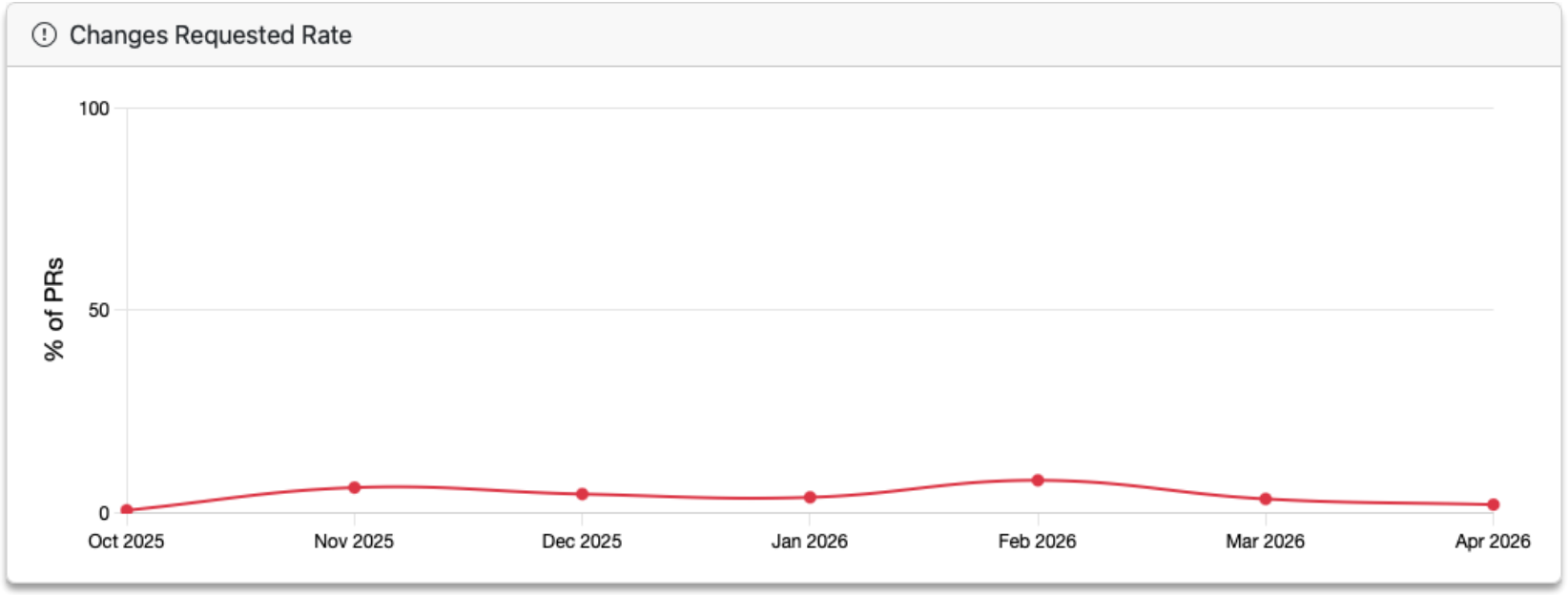
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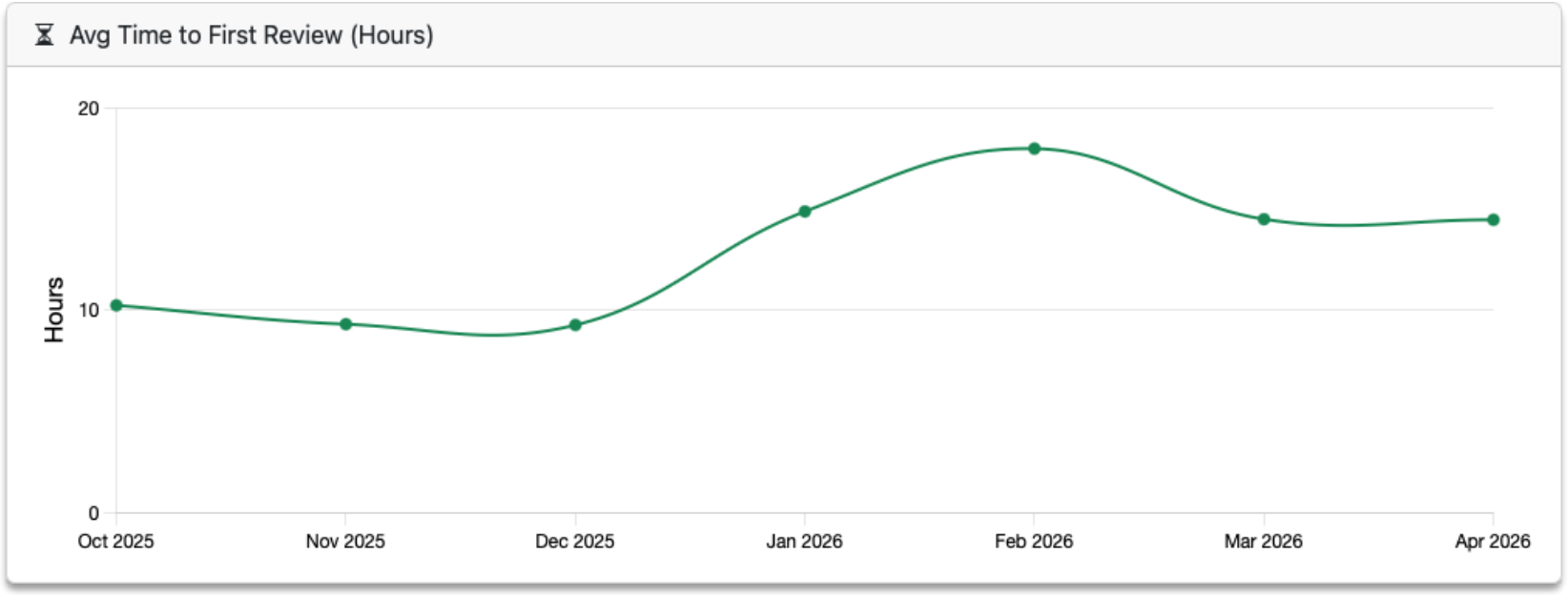
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04

AI-ASSISTED QUALITY ANALYSIS, RESPONSIBLY

Classification is a signal, not a verdict.

AI as your analytics engineer · not your judge

01

Classifies every PR comment

Bug? Missing test? Design issue? Praise?

02

Scores four dimensions

Testing, Code Quality, Collaboration, and Standards; weighted per your priorities.

03

Generates coaching summaries

Talking points for 1:1 prep; never verdicts, always questions.

04

Flags its own uncertainty

Every classification ships with a confidence score. Low confidence = verify.

How we prompt the AI

The classification prompt · simplified but faithful to what runs

```
SYSTEM
You are a code review analyst specializing
in PR comment classification.

Classify each comment into exactly these categories:
  bug_found | security_issue | missing_tests
  design_issue | dry_violation | style_issue
  praise | learning | nitpick | neutral ...

Return JSON: category, confidence (0.0-1.0),
reasoning.
```

Four dimensions · deliberately weighted

These are editorial choices. Pick weights that match YOUR team's values.

30%

TESTING

Testing habits have the highest leverage to improve.

25%

CODE QUALITY

Security issues, design patterns, first-pass approval rate.

25%

COLLABORATION

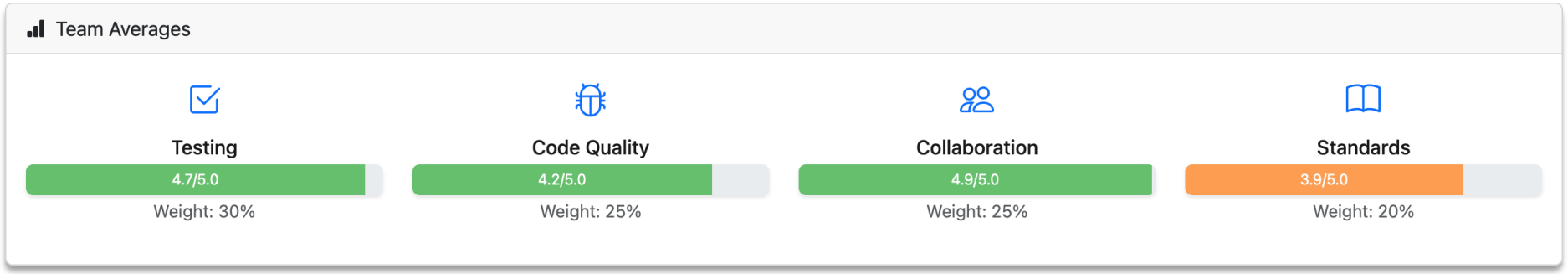
PR size, praise received, review engagement.

20%

STANDARDS

Design, DRY, indexing, style discipline.

Team averages



TESTING

4.7 / 5

CODE QUALITY

4.2 / 5

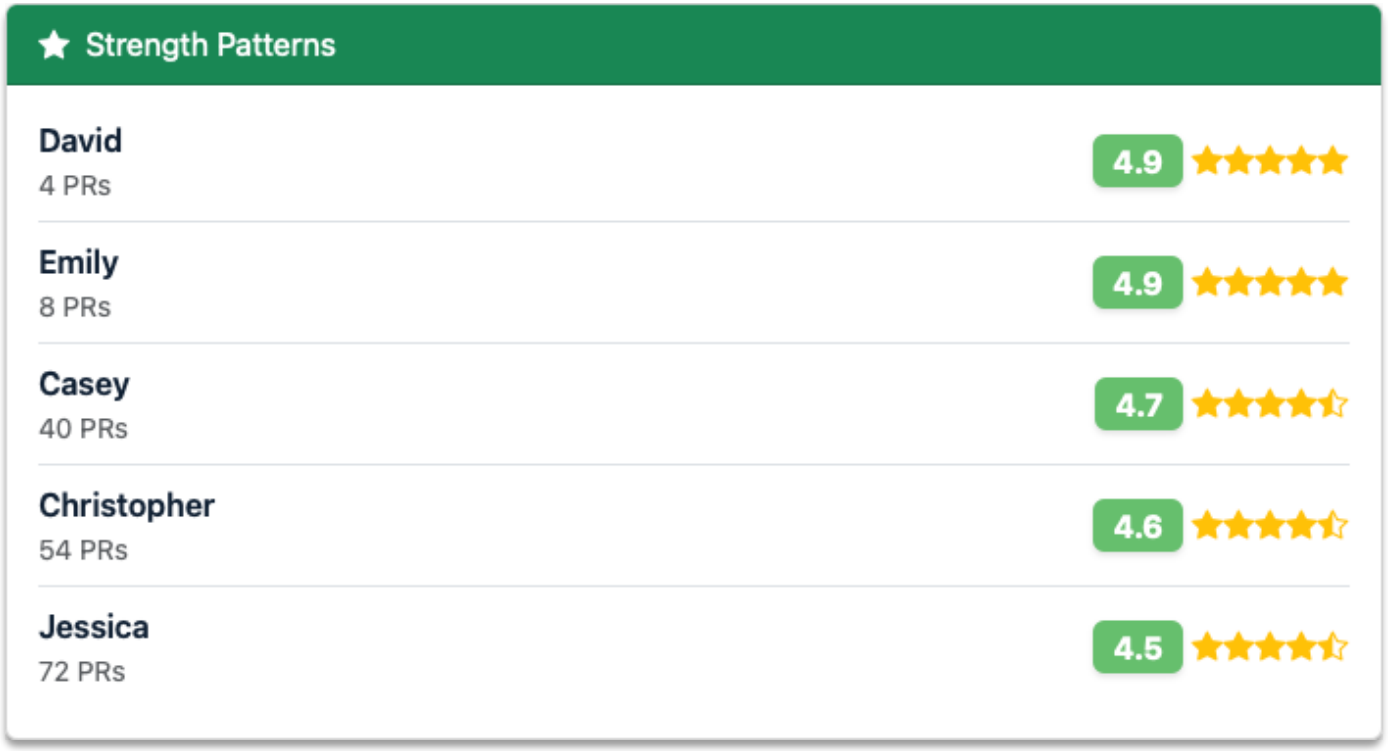
COLLABORATION

4.9 / 5

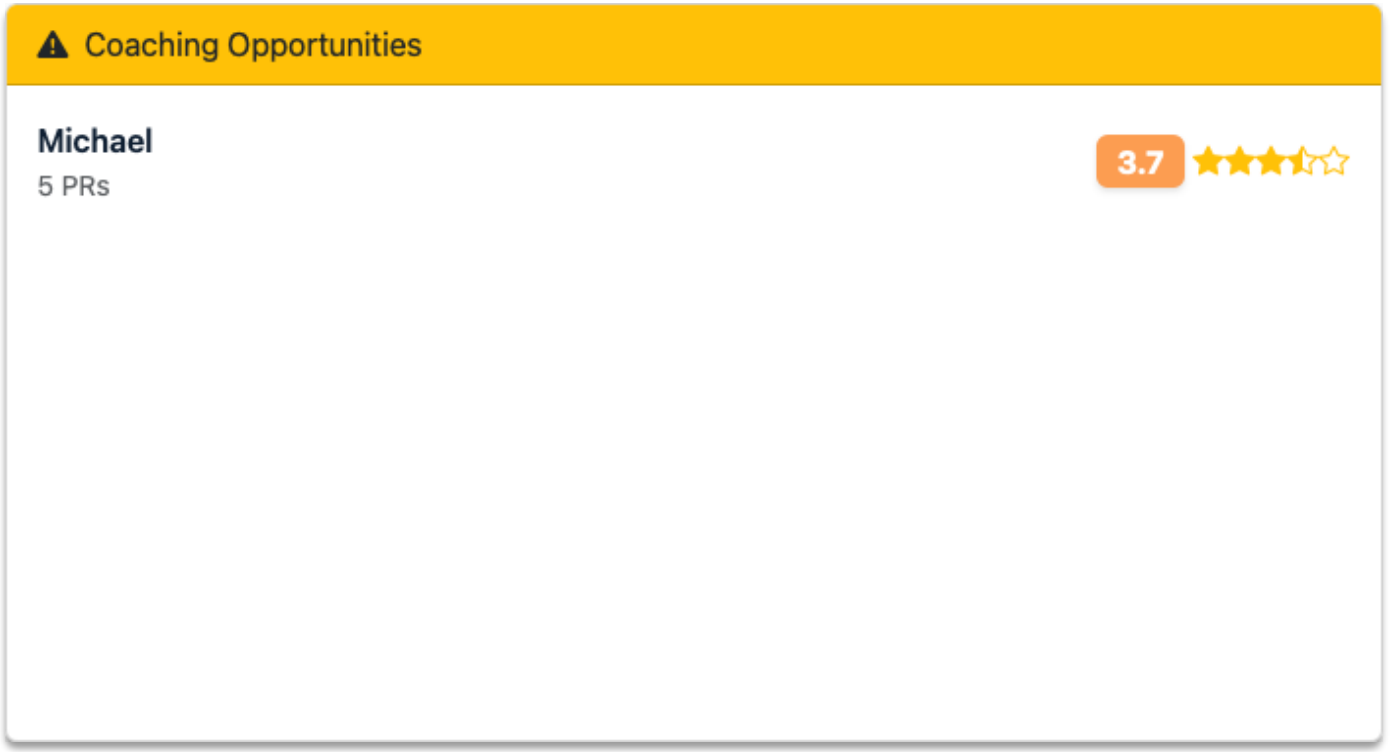
STANDARDS

3.9 / 5

Team ·Strength Patterns



Team ·Coaching Opportunities



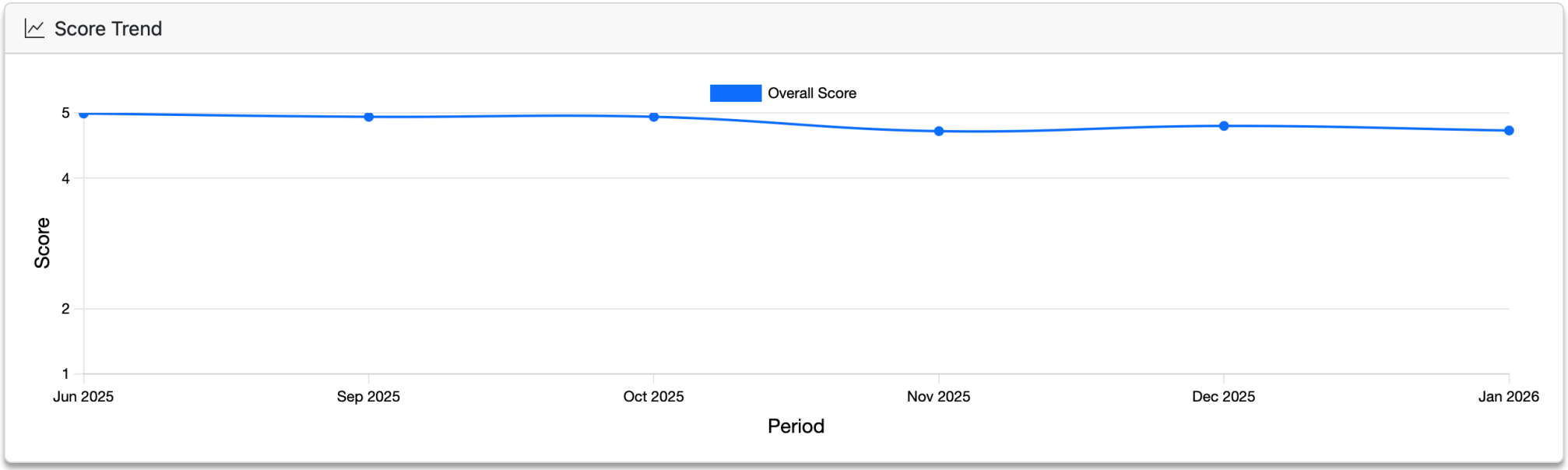
05

HOW TO APPLY THE SIGNALS

From notice → contextualize → conversation.

One engineer ·at a glance

Team context, PR signals, AI coaching suggestions; all on one page.



TEAM CONTEXT

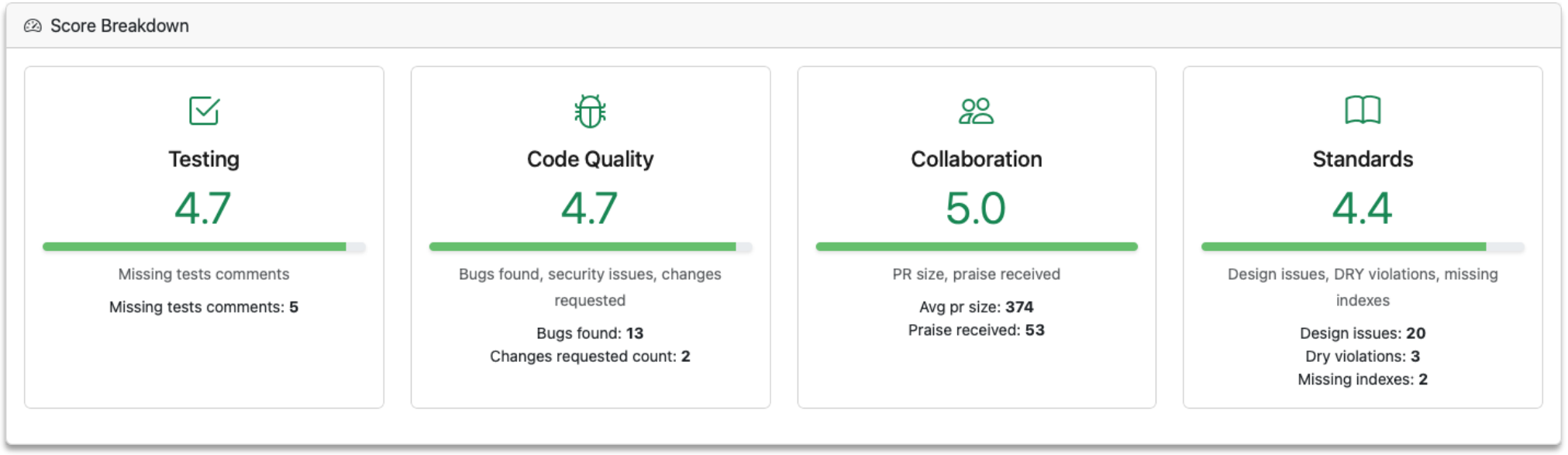
PR SIGNALS

COACHING PROMPT

SCORE BREAKDOWN

One engineer ·at a glance

Team context, PR signals, AI coaching suggestions; all on one page.



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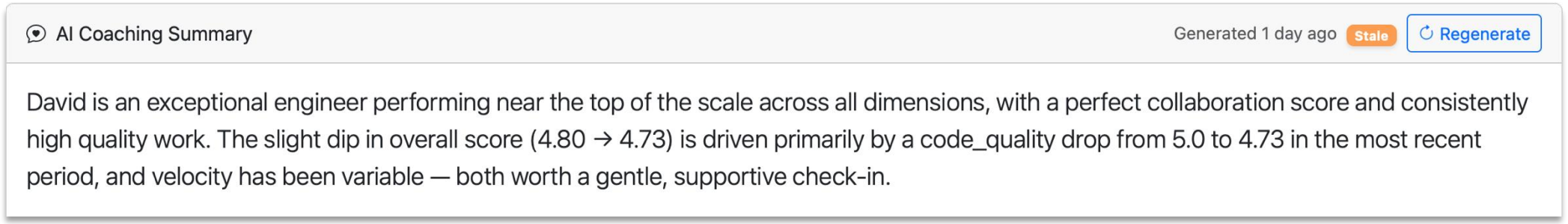
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SCORE BREAKDOWN

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★ Strengths

Collaboration

Perfect 5.0 collaboration score sustained across all 6 tracked periods, combined with 53 praise comments received on 44 PRs — that's more than one praise per PR on average.

💬 *"I want to call out that your collaboration is genuinely outstanding. Teammates are consistently praising your work — 53 praise comments across 44 PRs is remarkable. How do you think about the way you engage with others on reviews?"*

PR Approval Rate

42 out of 44 PRs approved with only 2 changes-requested, yielding a 95.5% approval rate. 30 first-pass approvals (68%) despite an average PR size of 374 lines.

💬 *"Your approval rate is excellent — 68% first-pass approval on PRs that average 374 lines shows you're shipping well-thought-out work. That's a strong signal of thoroughness before you even open the PR."*

Testing Discipline

Testing score of 4.72 held steady over the entire period, and only 5 missing-test flags across 44 PRs.

💬 *"Your testing habits are really consistent — only 5 missing-test flags across 44 PRs shows you're thinking about coverage proactively. That's a great example for the team."*

Peer Impact & Learning Culture

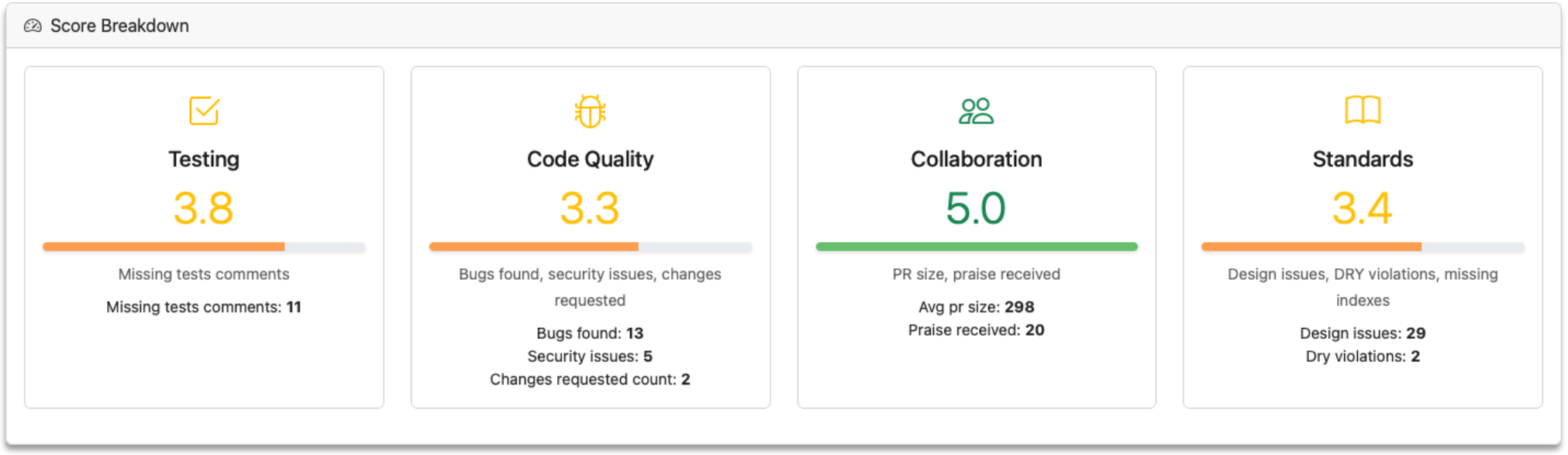
53 praise signals and 3 learning signals in classification highlights suggest David is both elevating others and growing himself.

💬 *"I noticed you're generating learning moments alongside all that praise — that combination tells me you're not just executing but also pushing yourself and helping others level up."*

TEAM CONTEXT **PR SIGNALS** **COACHING PROMPT** **SCORE BREAKDOWN**

One engineer ·at a glance

Team context, PR signals, AI coaching suggestions; all on one page.



TEAM CONTEXT

PR SIGNALS

COACHING PROMPT

SCORE BREAKDOWN

One engineer ·at a glance

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Growth Areas

Design Decisions in PRs

20 design_issue flags — the highest issue category — across 44 PRs means roughly 45% of PRs attracted at least one design concern. This is the single largest source of review feedback.

Suggestion: Consider investing more time in lightweight design docs or brief design discussions with a teammate before opening larger PRs. For PRs over 300 lines, a 10-minute pre-review conversation could reduce design churn significantly.

💬 *"Your code quality is really high, but I noticed design-related comments are your most common review feedback — 20 across this period. I don't think this means your designs are wrong; it might mean the reasoning isn't fully visible in the PR itself. Would it help to experiment with brief design sketches or pre-review chats on bigger changes?"*

Velocity Consistency

Points per cycle ranged from 2.7 to 24.0 over the last 6 cycles (average 11.5), with the most recent cycle at only 2.7. The velocity trend is -6.3, indicating a downward trajectory.

Suggestion: Explore whether the cycle 83 dip (2.7 points) reflects a natural rhythm (e.g., post-sprint recovery after the 24-point cycle 81), blocked work, or external factors. If it's work-in-progress carrying over, consider breaking work into smaller shippable increments.

💬 *"Your throughput has been quite variable — you had a huge 24-point cycle followed by a much lighter 2.7-point cycle. I'm not worried about the average; I'm more curious about what's driving the swings. Are there blockers or dependencies causing bunching, or is this just the natural shape of the work?"*

TEAM CONTEXT **PR SIGNALS** **COACHING PROMPT** **SCORE BREAKDOWN**

One engineer ·at a glance

Team context, PR signals, AI coaching suggestions; all on one page.

Recommended Actions

- Pair Michael with a senior engineer or architect for lightweight design reviews before she starts implementing larger features — this could directly reduce the 29 design issues surfacing in code review.
- Set a specific goal together: aim to increase first-pass approval rate from 27% to 40%+ next period by focusing on including tests and a brief design rationale in PR descriptions.
- Recognize her collaboration excellence publicly — her 5.0 score and high praise count make her a natural candidate for a team shoutout or a mentoring role for newer engineers.
- Explore whether the velocity spike in cycles 79 and 83 (30 and 33 points) correlated with any quality trade-offs. If not, understand what enabled those productive cycles and try to replicate those conditions.

Suggested Conversation Opener

"Michael, I wanted to start by calling out something that really stands out in your metrics — you have a perfect 5.0 collaboration score across every single period, and you've received 20 praise comments from teammates. That level of consistency in how you support the team is exceptional, and I want to make sure you know it doesn't go unnoticed."

TEAM CONTEXT

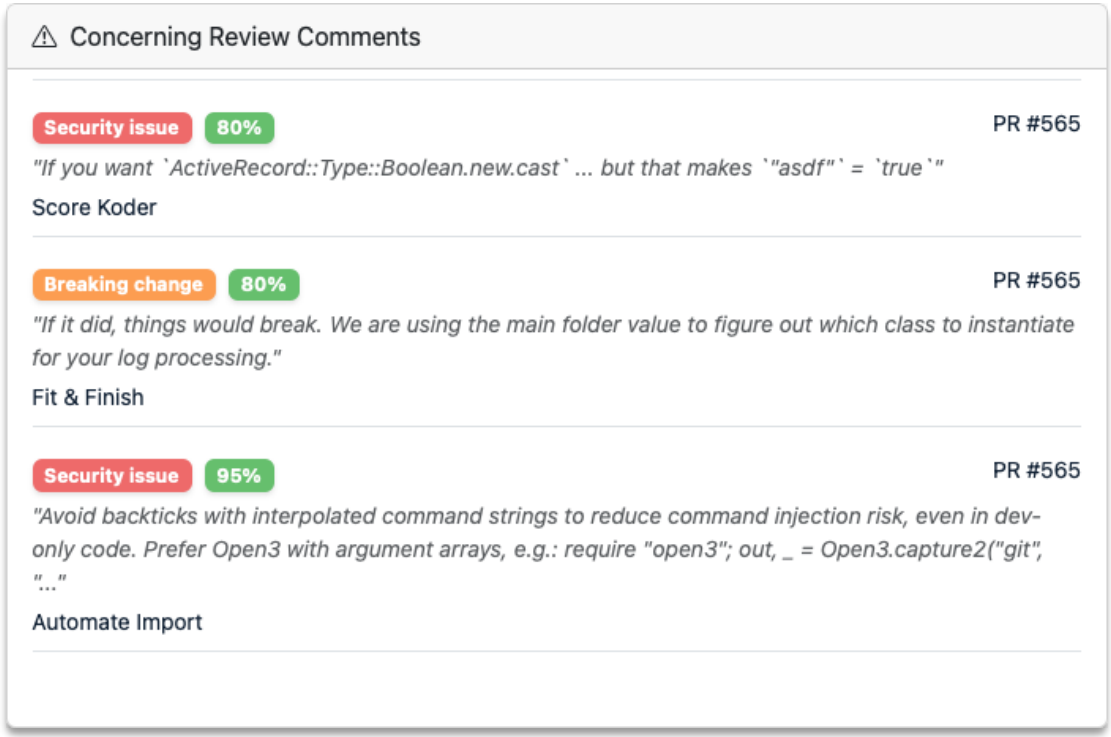
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TEAM CONTEXT

PR SIGNALS

COACHING PROMPT

SCORE BREAKDOWN

From numbers to conversation

AI-generated coaching summary with guardrails in the prompt.

```
You are an engineering leadership coach.

Analyze this engineer's metrics and generate
a coaching summary for their manager's 1:1.

Input: current scores, score history, PR metrics,
velocity data, classification highlights.

Return: overall assessment, strengths (with evidence),
growth areas (with talking points), velocity assessment,
recommended actions, conversation opener.

Guidelines:
- Frame growth areas as opportunities, not criticisms
- Reference actual numbers from the data
- If scores are improving, acknowledge the progress
```

From signal to action

Five steps to use the data without becoming THAT boss

1.

Notice

The dashboard flags a pattern; a declining score, creeping rework, a timing shift.

2.

Contextualize

Before you say anything, ask yourself if there's an obvious explanation.

3.

Discuss

Use the coaching summary talking points in your next 1:1, not the raw numbers.

4.

Support

Pair programming, targeted training, workload adjustment. This is where leadership happens.

5.

Track

Check the next cycle. Did support move the signal? Iterate if not.

Support first, not escalate. If your first instinct is a PIP, you've missed the point.

06

GUARDRAILS FOR TRUST

Without them, none of this works.

Guardrails that make this work

Privacy, transparency, automation; built in, not bolted on.

1. Presentation mode

One-click anonymization for demos and cross-team sharing

2. Engineers see their own data

Access-controlled, not hidden. Transparency breeds trust; hiding metrics breeds suspicion.

3. Including Human Relationships

Using real human communication in data and analytics

4. Confidence on everything

Every classification ships with a confidence score. Low confidence = manually verify.

5. Multi-dimensional Scoring & Trends

Can't game all metrics. We watch the slope, not the snapshot.

— If engineers don't trust the system, it doesn't matter how good the metrics are.

07

BUILD THIS YOURSELF

The stack is straightforward. The conversation with your team is the hard part.

5-phase build plan

Web App Framework, DB, background jobs, charting framework, and AI API

Phase 1

Connect data

Sync services for GitHub and Linear/Jira APIs.

Phase 2

Velocity math

Points, hours-per-point, chart dashboard.

Phase 3

PR analysis

Rework patterns, review cycle metrics.

Phase 4

AI classification

Comment categorization with AI.

Phase 5

Scoring & coaching

Confidence labels, coaching summaries.

Ongoing

Calibrate

Iterate and communicate

THE HARDEST PART

isn't the code.

It's the conversation with your team about why.

What to remember tomorrow morning

Three things to take with you.

3. **Guardrails are a feature.**
Credibility through transparency, confidence labels, and access control — with hours-per-point as your constant against time.

2. **Three tiers.**
Outcomes are the rearview mirror. System signals are the dashboard. Behaviors are the engine light.

1. **Start conversations, not end careers.**
If your first instinct when a metric dips is to escalate, you've missed the entire point.

Thank you

Metrics should start conversations, not end careers.

SCAN →



Let's connect.

Questions?

Eric Martin

Director of Software Engineering, KODE Health

linkedin.com/in/ericemartin1

GO HOME AND BUILD THIS YOURSELF

Now! Why are you still here?

Hours per point ·the great equalizer

TEAM A

50

points shipped
× 3 hrs/pt

= 150 hours

TEAM B

30

points shipped
× 5 hrs/pt

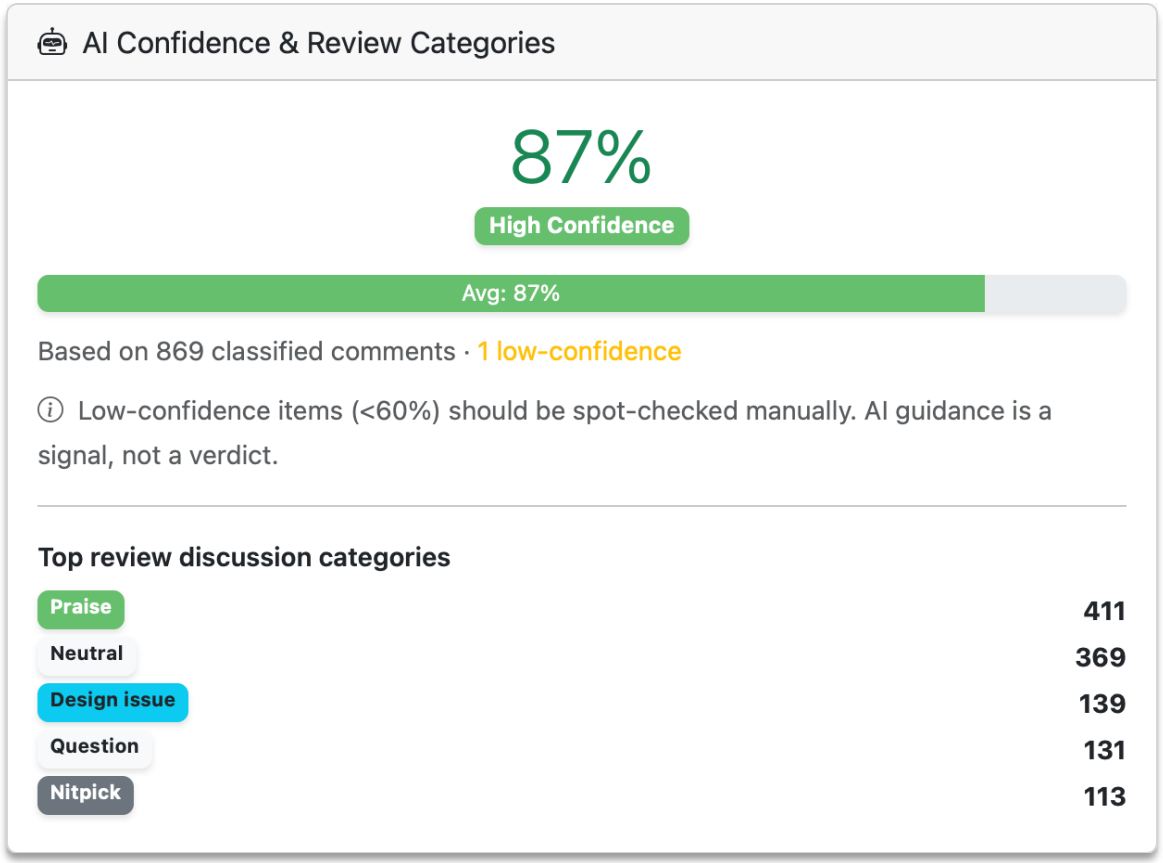
= 150 hours

— **SAME EFFORT. DIFFERENT CALIBRATION.**

Raw velocity hides this. Hours per point exposes it, and stops the "that team ships more" argument dead.

Trending up? Estimation is drifting. Trending down? Getting efficient, or cutting corners. Check quality.

Confidence


AI Confidence & Review Categories
87%
High Confidence
Avg: 87%
Based on 869 classified comments · 1 low-confidence
Low-confidence items (<60%) should be spot-checked manually. AI guidance is a signal, not a verdict.
Top review discussion categories
Praise 411
Neutral 369
Design issue 139
Question 131
Nitpick 113

Calibration is important

How to know if your AI is actually helping.

9,600+

COMMENTS CLASSIFIED

83%

AVG CONFIDENCE

HIGHEST CONFIDENCE

```
praise    — 89%
bug_found — 87%
```

LOWEST CONFIDENCE

```
neutral — 78%
nitpick — 78%
```

The AI classifies · The math scores

STAGE 01 · AI

Claude classifies comments.

INPUT

```
"This will crash in prod if X is nil"
```

↓

OUTPUT

```
bug_found  ·  confidence 0.87
```

STAGE 02 · MATH

Deterministic weights score.

WEIGHTS (PER OCCURRENCE)

```
security_issue      -3.0
missing_tests       -2.5
bug_found           -2.0
design_issue        -1.0
praise              +0.3
```

THE METRICS GRAVEYARD

Metrics to bury

What NOT to measure.

Lines of code

Congratulations! You just incentivized copy-paste. A senior who deletes 2,000 lines in a refactor looks unproductive. Every other measure says they're the MVP.

Individual velocity as a ranking

The moment velocity becomes a leaderboard; it stops measuring output and starts measuring gaming skill. Goodhart's Law in action.

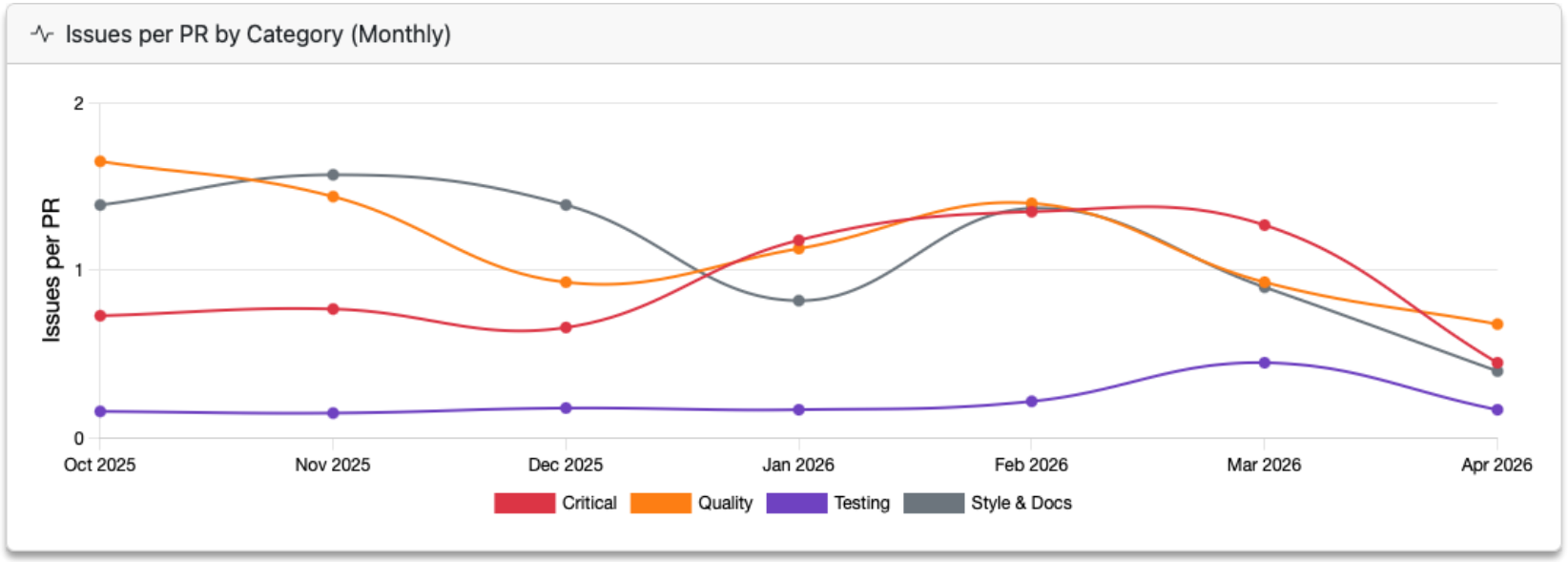
PR turnaround time, no context

Some PRs should take longer. A one-line typo fix and a 40-file migration are not the same PR. Treating them the same penalizes careful review.

Commit frequency

Your Git history is not a fitness tracker. This metric just teaches engineers that `git commit -m "wip"` fifteen times a day is what success looks like.

Standards drift · is the codebase getting better or worse?



CRITICAL **QUALITY** **TESTING** **STYLE & DOCS**

Each line = issue category, normalized per PR. Shipping more shouldn't spike the chart.